# Comparing ESM3 structures with experiments

HA and 2HMG, the neuraminidase test, and choosing the next target

8 November 2026

## 1 Short answers

**How do I save the structure ESM3 gives me?** With `write_pdb()` from `softmax/monte_carlo/decode_to_pdb.py`. It is our own helper, not an ESM function. The decode used everywhere in this repository, `decode_structure_with_mask()` in `compute_structural_metrics.py`, returns three tensors:

- `positions`, shape $(L, 37, 3)$: the coordinates of up to 37 atom types per residue (the "atom37" layout), in Å;
- `mask`, shape $(L, 37)$: which of those atoms exist;
- `plddt`, shape $(L)$: the confidence per residue, on 0–1.

`write_pdb()` turns these into a text `.pdb` file and puts 100×pLDDT in the B-factor column, where every viewer expects it. You normally do not call it yourself: the script `decode_to_pdb.py` decodes and saves in one command (Section 3). Section 4 shows the call from your own Python code.

**Is the trimer with glycine linkers in `references.py`?** No. `references.py` holds only single sequences (`zero_polymer`, `zero_polymer_mature`, the controls). `decode_trimer.py` builds the trimer on the fly from `zero_polymer`:

```
unit   = ref_seq[16:529]                # residues 17..529, 513 aa
trimer = ("G" * 25).join([unit] * 3)  # 3 x 513 + 2 x 25 = 1589 residues
```

`decode_to_pdb.py` now does the same with `--copies 3 --linker 25`. Note that `decode_trimer.py` takes a *Python slice* (`--residues 16:529`, 0-based, end excluded) while `decode_to_pdb.py` takes *residue numbers* (`--residues 17:529`, 1-based, both ends included). Both mean the same 513 residues.

The trimer file is written as three chains A, B, C, one per HA copy, each numbered 17–529 like the monomer. The glycine linkers are decoded (ESM3 sees the full 1589-residue sequence) but are *not* written to the file, since they are not part of HA; their mean pLDDT is printed in the log instead.

## 2 What the numbers mean

**RMSD** (root-mean-square deviation; this is the "wmd" your supervisor meant). Pair each residue's CA atom in the prediction with the same residue's CA in the experiment, rotate and shift the prediction onto the experiment as well as possible (the Kabsch algorithm finds the best rotation exactly), then

$$\mathrm{RMSD} = \sqrt{\tfrac{1}{N}\sum_{i=1}^{N} \|x_i^{\mathrm{pred}} - x_i^{\mathrm{exp}}\|^2}.$$

It is a typical distance in Å. About 1 Å: the same structure up to noise; 2–3 Å: same fold, some loops or a domain off; 4–8 Å: large parts placed differently; above about 10 Å for 500 residues: not the same shape. One misplaced region inflates it for the whole protein.

**Core RMSD** Fit, drop residues more than 2 Å off, refit, five times (what PyMOL's `align` reports by default). How well the matching part matches; always read it with the fraction of residues it kept.

**Within 1/2/4/8 Å** Fraction of residues within that distance after the global fit. "70% within 4 Å" is easier to read than an RMSD.

**TM-score** 0 to 1, independent of length. Above 0.5: same fold. Below about 0.17: like two unrelated structures.

**Per-chain RMSD** Each experimental chain (HA1, HA2, and for the trimer each copy) fitted on its own. Low per chain but high overall means the pieces are right but packed wrongly against each other.

**pLDDT** ESM3's confidence per residue: $> 90$ very confident, 70–90 confident, 50–70 low, $< 50$ roughly a guess. The script also gives Spearman(pLDDT, deviation); negative means the model is least sure exactly where it is wrong.

What to expect: HA decodes at mean pLDDT about 0.4–0.6 both as a monomer and as the G25 trimer (DEVLOG, 2026-11-04), so expect a large RMSD and a TM-score below 0.5. The comparison turns "the decoder does not fold HA" into a number against the real structure. Maltose-binding protein (Section 3.4) shows what a good match looks like.

# 3 Running it on astaroth

All commands are run from `softmax/monte_carlo`; adjust the first `cd` to your clone. Pull first so the scripts are there.

## 3.1 The experimental structure

```
cd ~/my_extendedSampling && git pull
cd softmax/monte_carlo
mkdir -p structure_compare
wget -O structure_compare/2HMG.pdb https://files.rcsb.org/download/2HMG.pdb
grep -c '^ATOM' structure_compare/2HMG.pdb   # thousands of lines = download worked
```

2HMG is the cleaved protein: separate HA1 and HA2 chains, three of each. The comparison script lists the chains it finds and which it used.

## 3.2 Predict and save (GPU)

Monomer (full HA0, the sequence the sampler works on, numbered 1–566):

```
nohup python -u decode_to_pdb.py --ref zero_polymer --num-steps 32 \
    --out structure_compare/ha0_pred.pdb > structure_compare/decode_ha0.log 2>&1 &
```

Trimer, exactly as `decode_trimer.py` builds it (ectodomain 17–529, three copies, 25 glycines between them):

```
nohup python -u decode_to_pdb.py --ref zero_polymer --residues 17:529 \
    --copies 3 --linker 25 --num-steps 32 \
    --out structure_compare/ha_trimer_pred.pdb \
    > structure_compare/decode_trimer.log 2>&1 &
```

Follow a log with `tail -f structure_compare/decode_trimer.log` (Ctrl-C stops watching, not the job). Each ends with `Saved -> ...`, after printing the mean pLDDT and radius of gyration (per chain and for the linkers, for the trimer). Run the two decodes one after the other if GPU memory is tight; the trimer is 1589 residues, the longest decode so far. `--num-steps` is the same setting as in the other decode scripts; HA's structure changes with it, so write the step count next to every RMSD you report.

### 3.3 Compare (CPU, seconds)

```
python compare_structures.py --pred structure_compare/ha0_pred.pdb \
    --exp structure_compare/2HMG.pdb --out-prefix structure_compare/ha0_vs_2HMG \
    | tee structure_compare/ha0_vs_2HMG.txt

python compare_structures.py --pred structure_compare/ha_trimer_pred.pdb \
    --exp structure_compare/2HMG.pdb --out-prefix structure_compare/trimer_vs_2HMG \
    | tee structure_compare/trimer_vs_2HMG.txt
```

How the script pairs residues:

- By sequence alignment, not by residue number, so the different numbering of our sequence and of 2HMG does not matter.
- **Monomer.** ESM3 gives one chain (HA0); 2HMG has HA1 and HA2 as separate chains. Both are aligned onto the one predicted chain. The other two copies in 2HMG would land on the same predicted residues, so they are skipped ("symmetry copy").
- **Trimer.** Residues already paired are hidden from the next alignment, so the three HA1/HA2 pairs of 2HMG land on the three predicted chains A, B, C. Which experimental copy goes with which predicted copy is arbitrary, so the script tries every order and keeps the lowest RMSD; the line "predicted protomers tried in every order" lists all of them. The trimer number therefore includes how the three copies are packed, the thing the trimer was built to test.
- To compare one predicted copy only: `--pred-chain A`. To choose experimental chains by hand: `--exp-chains A,B`.
- Residues absent from the crystal (signal peptide, transmembrane anchor, tail, disordered loops) cannot be compared and are left out of every number; the script says how many.

Files written next to the prefix:

| | |
|---|---|
| `_superposed.pdb` | the prediction moved on top of 2HMG |
| `_per_residue.csv` | per residue: predicted chain and number, pLDDT, matching 2HMG chain and number, deviation |
| `_per_residue.png` | deviation (blue) and pLDDT (red) along the file; dashed lines separate experimental chains |
| `_view.cxc / .pml` | ready-made ChimeraX / PyMOL scenes |

### 3.4 Positive control

Maltose-binding protein decodes at pLDDT 0.93; PDB 1ANF is its crystal structure.

```
wget -O structure_compare/1ANF.pdb https://files.rcsb.org/download/1ANF.pdb
nohup python -u decode_to_pdb.py --uniprot P0AEX9 --num-steps 32 \
    --out structure_compare/mbp_pred.pdb > structure_compare/decode_mbp.log 2>&1 &
python compare_structures.py --pred structure_compare/mbp_pred.pdb \
    --exp structure_compare/1ANF.pdb --out-prefix structure_compare/mbp_vs_1ANF
```

`--uniprot` needs network access to UniProt and takes the mature chain.

## 4 Calling it from your own Python code

If you already have the tensors from a decode inside your own script (for example a sampled sequence), save them like this. Run from `softmax/monte_carlo` so the imports resolve, as in the other scripts.

```
from compute_structural_metrics import decode_structure_with_mask
from decode_to_pdb import write_pdb
from custom_esm.utils.residue_constants import atom_types

positions, mask, plddt = decode_structure_with_mask(sampler.model, eprot, config)

write_pdb("my_structure.pdb", eprot.sequence,
          positions.float().numpy(), mask.bool().numpy(),
          plddt.float().numpy() * 100,     # pLDDT 0-100 into the B-factor column
          atom_types)
```

For a G-linked trimer, pass `segments` so each copy becomes its own chain and the linkers are left out. Each entry is (chain, first index, end index, first residue number), with 0-based indices into the decoded sequence:

```
n, L = 513, 25                          # residues per copy, glycines per linker
segments = [("A", 0,           n,           17),
            ("B", n + L,       2 * n + L,   17),
            ("C", 2 * (n + L), 3 * n + 2*L, 17)]
write_pdb("trimer.pdb", eprot.sequence,
          positions.float().numpy(), mask.bool().numpy(),
          plddt.float().numpy() * 100, atom_types, segments=segments)
```

ESM's own `ESMProtein.to_pdb()` also writes PDB files, but the decode path here never builds an `ESMProtein` with coordinates, and the helper the repository used before (`test_cleaved_complex.py --pdb`) leaves every B-factor at 1.00 and writes pLDDT to a separate `.plddt.tsv`. `write_pdb()` keeps everything in one file.

## 5 Looking at the structures

A viewer draws 3D graphics, so it needs a graphical session on the machine that runs it. astaroth has a screen, so pick by where you are sitting.

**At astaroth's keyboard** install the viewer there and open the files in place. Nothing to copy.

**Over ssh** either render the picture on astaroth and copy back only the image, which needs no graphics session at all,

```
chimerax --offscreen --exit --cmd "open trimer_vs_2HMG_view.cxc ; \
    save view.png width 2000 supersample 3"
```

(`-offscreen` needs a build with OSMesa, which the Linux builds normally have; if yours does not it says so immediately), or copy the folder to your laptop and look at it there (run this from the laptop; replace the login and path):

```
scp -r marco.sanchini@astaroth:~/my_project/extendedSampling/softmax/monte_carlo/
    structure_compare .
cd structure_compare
```

Copy the whole folder, not single files: the `.cxc` and `.pml` scripts refer to the others by relative path. `ssh -X` also works but forwards every frame, so it is only good for checking that a file opens.

**ChimeraX** (free for academic use, `https://www.cgl.ucsf.edu/chimerax/download.html`). Quickest: *File > Open...* and pick `trimer_vs_2HMG_view.cxc`. 2HMG appears in grey, the prediction coloured by pLDDT (dark blue > 90, light blue 70–90, yellow 50–70, orange < 50), already on top of each other. By hand, in the command line at the bottom of the window:

```
open 2HMG.pdb                          # model #1
open ha_trimer_pred.pdb                # model #2
delete solvent
matchmaker #2/A to #1/A                # superimpose; prints an RMSD in the log
color #1 lightgray
color bfactor #2 palette alphafold
view
```

`matchmaker` fits one chain to one chain (here predicted copy A onto HA1 chain A), so its RMSD differs from the script's, which fits everything at once. It prints two values: over "pruned" pairs (like the core RMSD) and "across all pairs". `save picture.png width 2000 supersample 3` saves an image.

**PyMOL** (`conda install -c conda-forge pymol-open-source`). *File > Run Script...* and pick `trimer_vs_2HMG_view.pml`, or type:

```
load 2HMG.pdb, exp
load ha_trimer_pred.pdb, pred
remove solvent
align pred and name CA, exp and name CA, cycles=0   # all residues, no outlier
    rejection
hide everything
show cartoon
color grey70, exp
spectrum b, red_white_blue, pred, minimum=30, maximum=90
orient
```

The log line `Executive: RMSD = ...` gives the RMSD.

**Browser, nothing to install.** `https://molstar.org/viewer/`: drag in `2HMG.pdb` and the `_superposed.pdb` file (Mol* does not fit them for you), then set the prediction's colour theme to "Uncertainty/Disorder" to colour by the B-factor column.

# 6 Things that trip people up

- **Slice versus residue numbers**: `16:529` in `decode_trimer.py` is `17:529` in `decode_to_pdb.py`.
- **One chain against two**: HA0 is one chain in the prediction, HA1 + HA2 in the crystal. The script handles it; single-chain tools such as ChimeraX's `matchmaker` give different numbers.
- **Monomer against trimer**: a monomer prediction has no partner copies holding the stalk helices, so part of its disagreement with 2HMG is quaternary. Compare the per-chain RMSDs of the monomer and trimer runs.
- **Step count**: report `--num-steps` with every number.

- **Strain**: the identity line says how close 2HMG's sequence is to ours; mention it if it is well below 100%.

## 7 The neuraminidase candidate

HA0 failed the decode gate, so the sampling target is being chosen again (`target_selection.md`). The first candidate is the influenza neuraminidase catalytic head, and unlike HA it comes with an experimental structure of the construct itself: 2HU0 chain A, supplied by the supervisor stripped to one chain, kept at `code/bridge_test/pdb/na_head_2HU0_monomer.pdb`. Its sequence is `references.py` key `na_head_2hu0`, 385 aa.

Everything that made HA awkward is absent here: one predicted chain against one experimental chain, no cleavage, no trimer, and no waters or ligands left in the file, so nothing has to be stripped and `matchmaker #2 to #1` needs no chain selector.

```
python decode_to_pdb.py --ref na_head_2hu0 --num-steps 32 \
    --out structure_compare/na_head_pred.pdb

python compare_structures.py \
    --pred structure_compare/na_head_pred.pdb \
    --exp ../code/bridge_test/pdb/na_head_2HU0_monomer.pdb \
    --out-prefix structure_compare/na_head_vs_2HU0
```

How to read it. A mean pLDDT near 0.9 with a low RMSD means the candidate passes the gate and sampling can start on it; a pLDDT near HA0's 0.5 means drop it and move to the next candidate. The picture carries information the numbers do not: a six-bladed $\beta$-propeller is a pinwheel, so rotate until you look down the central axis and six four-stranded sheets should fan out around it. A prediction that is compact but shows no pinwheel has failed even if its RMSD is not catastrophic. Run `--num-steps 8` and `128` as well, to different `--out` names: a confident decode barely moves between step counts, while HA0 wandered from $R_g$ 45 to 70 Å.

Traps specific to this pair, on top of the list above:

- **Numbering**: the crystal runs 83–468 with five gaps totalling six unobserved loop residues and insertion codes 169A and 412A–D, while the prediction runs 1–385. The offset is not a constant 82 and changes at every gap, so take the pairing from `na_head_vs_2HU0_per_residue.csv` and never compute it by hand.
- **The reference is the observed sequence**: those six residues are missing from `na_head_2hu0`, which closes the gaps up. Acceptable for the gate, since all six are in loops, but before any sampling run or K-site comparison replace it from 2HU0's SEQRES (`pdb_to_reference.py --use-seqres` on the unstripped file) or from the strain's UniProt entry (`add_reference.py`).
- **A tetramer, but not HA's problem**: native NA is a homotetramer with a $Ca^{2+}$ site on the fourfold axis, so a monomer prediction is missing partners here too. The difference is that the head's fold is entirely intra-chain, a self-contained $\beta$-propeller, whereas HA2's long helix exists only as part of the trimeric coiled coil. Expect the interface surface to be the loosest part of the prediction, and flag interface residues before the per-site entropy comparison: they are conserved for quaternary reasons a single-chain energy cannot see.

# 8 Summary for the meeting of 9 November

Everything from here on was written for the supervisor meeting. In one page:

- **Neuraminidase does not pass the decode gate.** Mean pLDDT 68, C$\alpha$ RMSD 14.9 Å against the crystal your supervisor supplied, and only 39% of the crystal's contacts recovered. Repeating the decode at 8, 32 and 128 generation steps gives the *same* structure, so this is not a sampling artefact (section 9).
- **But the failure is specific, not diffuse.** A 116-residue core superposes at 1.17 Å, the overall size is right, and the one long-range disulfide bridge of the protein — Cys92 to Cys417, 325 residues apart — is placed 20.5 Å wide instead of 4.1 Å. The prediction looks like a six-bladed propeller whose ring never closes (sections 9.1 and 10).
- **Cytochrome P450cam passes, comfortably.** Mean pLDDT 92, RMSD 1.98 Å over 405 residues, core 0.96 Å over 79% of the chain, TM-score 0.94, 88% of the crystal's contacts recovered (section 12). It is 414 residues, monomeric, soluble, and from a very large sequence family, so it satisfies every criterion we set.
- **A measured threshold, instead of a guessed one.** Expressing the results in the contact-map energy of equation (1) gives $E = 0.11$ for P450cam, $E = 0.18$ for protein G (the paper's own target, which is predicted well but is short) and $E = 0.60$ for neuraminidase. So "good" is $E \approx 0.1$–$0.2$ and "failed" is above 0.5, with the caveat that short chains sit higher (section 12.1).
- **One thing to be clear about with the supervisor.** The energy the sampler actually minimises never looks at a crystal structure: it compares the model's output for a candidate sequence with the model's output for the *reference sequence*, so it is zero at the reference by construction. The decode gate therefore does not test whether the sampler is well posed; it tests whether its reference means the real protein (section 11). This is the point on which "can we use neuraminidase anyway" turns.
- **The $\beta$-propeller fold is not the problem.** The control was run on 8 November: a monomeric bacterial sialidase (NanH, 2SIL, 381 residues), which has the *same* six-bladed propeller fold as neuraminidase, is predicted well — pLDDT 86, core 0.94 Å over 73% of the chain, TM 0.895, and its C-terminal block lands at 0.83 Å, so the ring closes. Whatever defeats neuraminidase is specific to neuraminidase (section 13).
- **The viewer check is done** (9 November, figure 1): the crystal shows the closed six-bladed ring, the prediction builds blade-like sheets but not the ring. Consistent with the open-ring reading, though not on its own conclusive.
- **Cheapest remaining experiments**: re-decode NA with its disulfides supplied as constraints, using the tooling already in `code/bridge_test` (an hour); and, if the quaternary explanation is to be settled, a real multimer predictor on the tetramer (a week).

# 9 What the neuraminidase test measured

Section 7 said how to run the gate. This section is the result, run on astaroth on 6 November 2026. The decode was repeated at three generation step counts, because step-count instability was HA0's tell.

| steps | mean pLDDT | $R_g$ (Å) | RMSD, all (Å) | core RMSD / residues | TM |
|---|---|---|---|---|---|
| 8 | 65.8 | 19.62 | 13.59 | 1.17 / 117 (30%) | 0.592 |
| 32 | 67.7 | 19.72 | 14.93 | 1.17 / 116 (30%) | 0.563 |
| 128 | 74.0 | 19.49 | 13.17 | 1.11 / 120 (31%) | 0.583 |
| experimental, 2HU0 chain A | | 18.88 | — | — | — |

Three things follow immediately.

**It is not under-sampling.** Over a sixteen-fold range of step counts the geometry does not move: $R_g$ within 0.23 Å, the core fixed at 1.1–1.2 Å over 116–120 residues, TM 0.56–0.59, global RMSD 13–15 Å with no trend. The three runs' per-residue error profiles correlate with each other at 0.83–0.90, so the same residues are wrong by the same amounts every time. The decoder converges on one specific structure that is about a third correct.

**More steps buy confidence, not accuracy.** pLDDT rises from 65.8 to 74.0 and the fraction of residues below 50 falls from 26% to 5%, while the error stays where it was. A model that is uncertain and wrong can be caught by its own confidence; a model that is confident and wrong cannot.

**The error is localised, and not only at the chain ends.** By blocks of 20 residues at 32 steps: 1–20 at 29.6 Å, 101–240 at 2.8–4.7 Å, 241–300 at 7.7–14.8 Å, 301–340 at 15.5–32.6 Å, 341–385 at 21.5–41.1 Å. Trimming 30 residues from each end still leaves 12.1 Å over the remaining 325, so this is not a pair of floppy tails.

### 9.1 Two further diagnostics

`prediction_diagnostics.py` (new, 8 November) adds two numbers an RMSD does not give:

```
python prediction_diagnostics.py --pred structure_compare/na_head_pred.pdb \
    --exp ../code/bridge_test/pdb/na_head_2HU0_monomer.pdb \
    --csv structure_compare/na_head_vs_2HU0_per_residue.csv
```

**The contact-map energy.** The effective energy of Zambon et al. (2024) is not an RMSD but a normalised difference of contact maps,

$$E = \frac{\sum_{ij} |\Delta_{ij}(r_1) - \Delta_{ij}(r_2)|}{\sum_{ij} [\Delta_{ij}(r_1) + \Delta_{ij}(r_2)]}, \qquad |i-j| > 2, \tag{1}$$

where $\Delta_{ij} = 1$ when residues $i$ and $j$ are in contact and $E = 0$ means identical maps. Evaluating it between the *predicted* structure and the *crystal* asks, in the natural structural units of this project, how far the model's idea of neuraminidase is from the real protein. With a C$\alpha$–C$\alpha$ cutoff of 8 Å (the coarse stand-in for the paper's 4 Å heavy-atom definition, since our decodes carry backbone and C$\beta$ only):

| | contacts predicted | in crystal | shared | $E$ |
|---|---|---|---|---|
| NA head, 32 steps | 1312 | 1403 | 549 | 0.596 |
| NA head, 128 steps | 1386 | 1403 | 526 | 0.623 |

Only 39% and 37% of the crystal's contacts are reproduced. Note that more generation steps make this slightly *worse* even as pLDDT improves, which is the same pattern as above.

**Disulfide bridges, which say *where* the assembly failed.** The crystal has eight disulfide bridges in this single chain, found as cysteine pairs whose S$\gamma$ atoms lie within 2.5 Å. Measuring the same pairs in the prediction as C$\beta$–C$\beta$ distances, where a real bridge is 3.8–4.1 Å:

| pair (2HU0 numbering) | residues apart | crystal (Å) | predicted (Å) |
|---|---|---|---|
| 92–417 | 325 | 4.10 | **20.54** |
| 124–129 | 5 | 4.05 | 3.29 |
| 183–230 | 47 | 3.78 | 4.07 |
| 232–237 | 5 | 4.05 | 5.52 |
| 278–291 | 13 | 3.95 | 4.04 |
| 280–289 | 9 | 3.95 | 4.14 |
| 318–336 | 17 | 4.20 | **8.64** |
| 421–447 | 25 | 4.13 | **10.43** |

Five of eight are formed. The one that fails worst, by a factor of five, is the only long-range one: Cys92 to Cys417, 325 residues apart, the bridge that staples the beginning of the chain to its end. The 128-step run behaves the same way (15.7 Å). So the prediction gets the local covalent topology right and the one global staple wrong, which is a far more specific statement than "RMSD 14 Å".

# 10 Why the neuraminidase prediction might fail

Several explanations fit "pLDDT 68, one third correct". They have different consequences and most can be separated by a cheap experiment. Each is stated below with the evidence we have for and against it and with the test that would settle it. None is established; that is the point of listing them.

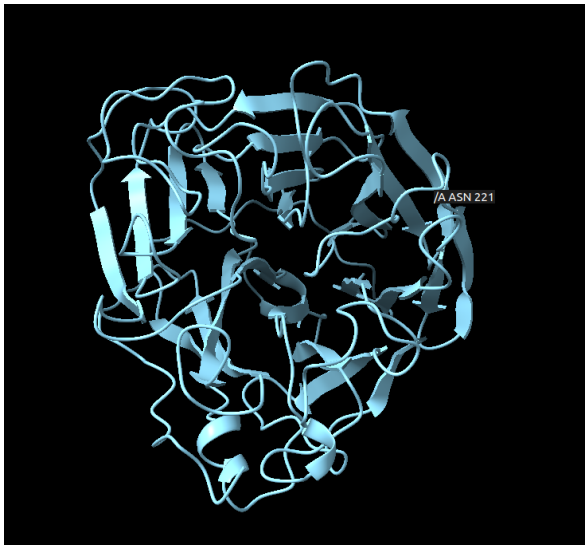
## 10.1 The propeller ring does not close

**The claim.** The catalytic head is a six-bladed $\beta$-propeller: six repeats ("blades"), each a four-stranded antiparallel $\beta$-sheet, arranged around a central axis like turbine vanes, with the active site in the funnel on the axis (Varghese, Laver & Colman, 1983). The chain threads the blades in order, so its first and last strands end up side by side in the *same* blade, and that blade is complete only because the C-terminal strand packs against the N-terminal ones. The closure is called the molecular Velcro and it is what makes the arrangement a closed ring rather than an open spiral (Fülöp & Jones, 1999). A predictor that builds most blades but never makes that last contact leaves the ring open, and the two ends of the chain, which should be pressed together, drift apart.

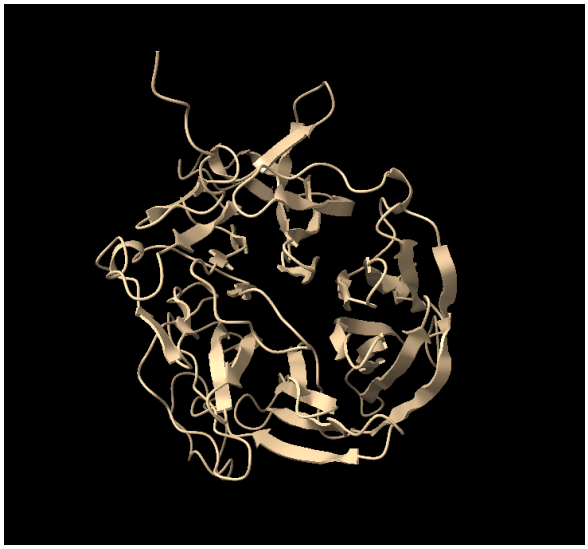
**For.** The best-supported explanation we have. The middle of the chain is 3–5 Å while *both* ends are 20–40 Å out, which is the signature of an unclosed ring rather than of a generally poor model. And Cys92–Cys417, the covalent staple between the chain's beginning and its end, is the single worst-predicted feature in the structure (section 9.1).

**Against.** Nothing that the pictures contradict, but see the caveat at the end of this subsection: the viewer check supports this reading without proving it.

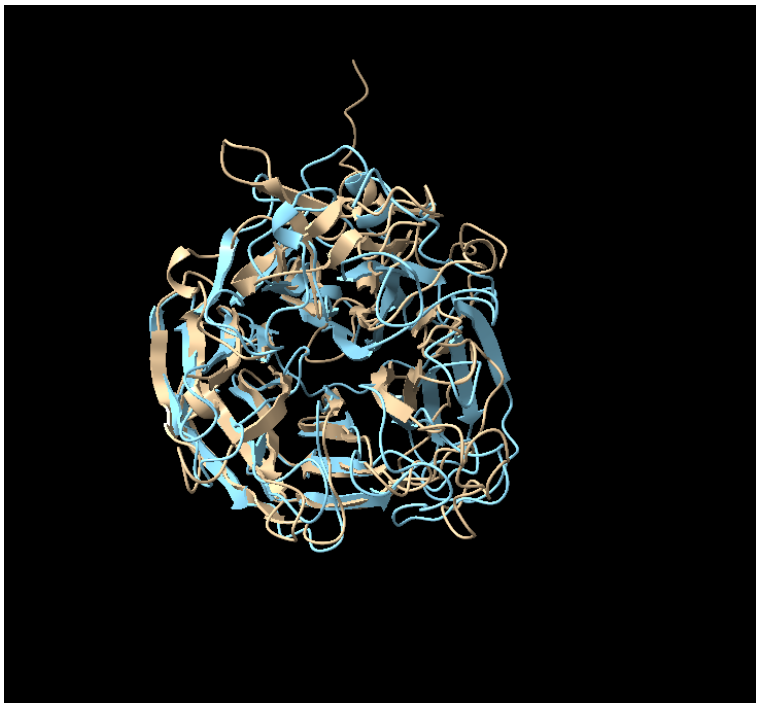
**Test, and the result (9 November).** The test was thirty seconds in ChimeraX: open the two structures, look down the propeller axis, count the blades, and see whether the first and last strands sit in one blade or point away from each other. Figure 1 is what came out.




(a) crystal, 2HU0 monomer



(b) ESM3 prediction, 32 steps



(c) the two superposed (ChimeraX `matchmaker`), crystal blue, prediction beige

Figure 1: Looking down the pseudo-sixfold axis. In the crystal (a) the six wedge-shaped blades fan out from a central point, each an antiparallel $\beta$-sheet, and the ring is closed: following the outside of the molecule never leaves the sheet. The prediction (b) has a comparable diameter and plenty of $\beta$ structure, but the radial organisation survives only in part of the molecule; elsewhere the sheets are short or replaced by loops, there is no clean central funnel, and a long unstructured segment — the N-terminus, the block measured at 29.6 Å in section 9 — leaves the body of the protein at the top. The superposition (c) shows that the two occupy a similar envelope and that many individual segments run parallel, which is the 1.1 Å core being matched, while the rest sits beside the crystal rather than on it.

**What the pictures do and do not establish.** They are consistent with an unclosed ring and with nothing else we have proposed: the local blade motif is clearly being built, and it is the global arrangement that is wrong. But the honest limit is that a ribbon drawing at this size does not let one count six blades reliably or point to the strand that should complete the first one, and `matchmaker` aligns on the best-matching subset, so a respectable-looking overlay is exactly what a partially correct model gives. Taken with the numbers — both chain ends 20–40 Å out while the middle is 3–5 Å, and Cys92–Cys417 at 20.5 Å instead of 4.1 Å — the picture is supporting evidence, not a proof. The measurement that would settle it is the disulfide-constrained re-decode below.

### 10.2 The blades are built in the wrong register

**The claim.** Six blades of one fold are six near-identical sequence repeats. A model that recognises "propeller" but misassigns which repeat goes where would produce locally correct blades in globally shifted positions; a shift by one blade would leave local geometry excellent and swap the chain ends.

**For.** Locally the model is very good — 1.1 Å over 30% of the chain — which is what a register error looks like. Repeat proteins are a known hard case for structure predictors.

**Against.** A clean one-blade shift should leave a larger correct core than 30%.

**Test.** Superpose the prediction's blade $k$ on the crystal's blade $k \pm 1$ and see whether a shifted assignment fits better than the straight one. Needs blade boundaries from the crystal's secondary structure: an hour, not minutes.

### 10.3 Neuraminidase is an obligate tetramer

**The claim.** The native protein is a homotetramer with a $Ca^{2+}$ ion on the molecular fourfold axis, and the head's surface includes a large interface that in life is never exposed to solvent. A predictor given one chain must fold a surface that does not exist on its own.

**For.** HA failed for the analogous reason, and this is the obvious prior for a viral surface glycoprotein. It is also now the *leading* explanation, because the control of section 13 removed the alternative that propellers as such are hard: the protein that succeeds is monomeric and the one that fails is an obligate tetramer, which is the most visible difference between them.

**Against.** Unlike HA2's long helix, the head's fold is entirely intra-chain: every contact defining the propeller is within one chain, and isolated heads are used experimentally. This is why NA was the better candidate in the first place.

**Test.** Two. (i) Map the badly predicted residues onto the tetramer interface, computed from 2HU0's full biological assembly: if the error sits on the interface the claim gains, if it sits in the ring closure it does not. The unstripped entry has to be downloaded on astaroth, since the cloud container cannot reach RCSB. (ii) Predict the tetramer with a real multimer predictor (AlphaFold-Multimer, Boltz, Chai): useful to know, a week of work, and not needed for the target decision.

### 10.4 The fold depends on disulfides the model does not know about

**The claim.** Eight bridges in 385 residues is a lot, and in the real protein they form in the endoplasmic reticulum *during* folding. A sequence-only predictor has no explicit notion of them.

**For.** Three of eight are not formed (section 9.1).

**Against.** Cause and effect are hard to separate: a wrong global fold necessarily misplaces a long-range bridge, so the missing bridge may be a symptom rather than the cause.

**Test.** The tooling already exists here: the bridge experiments in `code/bridge_test` prompt ESM3 with known disulfides (`*_prompt_allbonds_*`), developed for BPTI and crambin. Supplying the eight bridges as constraints and re-measuring separates the two readings: if the fold snaps into place, the model's problem is that it cannot *find* the bridges, not that it does not know the fold. The cheapest informative experiment after the viewer check.

### 10.5 Glycosylation and ligands are absent

**The claim.** Neuraminidase is a glycoprotein with several N-linked glycosylation sequons, and the crystal has a $Ca^{2+}$ site. The prediction is a bare polypeptide.

**For.** Loops held in place by a glycan or an ion are exactly the local features a bare prediction gets wrong.

**Against.** Glycans and ions rarely determine a *global* fold, and the error here is global. A contributing factor at most.

**Test.** Check whether the displaced residues coincide with the sequons and the metal site; cheap once the interface analysis is set up.

### 10.6 The construct is a fragment

**The claim.** Residue 83 is an artificial chain start: in the real protein the stalk continues towards the membrane, so the model may be folding an N-terminus that does not exist in life.

**For.** The first 20 residues are the worst block at 8 and 32 steps.

**Against.** At 128 steps the largest deviations move off the N-terminus onto residues 430–437, and residue 468 is near the real C-terminus, so this does not cover the 301–385 region at all.

**Test.** Decode the head plus the stalk, or the full-length sequence, and see whether the head improves. One decode.

### 10.7 The model is simply weaker on this class of fold

**The claim.** The dullest explanation, and therefore the one to exclude first: ESM3-open, sequence only and without an MSA, may be worse on $\beta$-rich repeat folds than on the compact $\alpha/\beta$ globular proteins we happen to have calibrated on. Our passing controls — maltose binding protein, amylase, luciferase, albumin, and now P450cam — are all of the latter kind.

**For.** It would explain everything we see without invoking neuraminidase at all.

**Against. This explanation is now excluded**, by the control described in section 13: a monomeric bacterial sialidase with the same fold is predicted well. ESM3 can build a closed six-bladed propeller.

**Test, and it is a good one.** Bacterial sialidases have the *same* six-bladed propeller fold as viral neuraminidase but are monomeric; the classic case is the sialidase of *Salmonella typhimurium* (NanH, PDB 2SIL, about 380 residues), whose structure was solved precisely to demonstrate the shared fold (Crennell et al., 1993). One decode separates the hypotheses:

- NanH folds well ⇒ the propeller fold is not the problem, and what fails in NA is specific to NA: the tetramer, the glycans, or this particular ring closure.
- NanH fails too ⇒ the problem is the fold class. Our tooling cannot be trusted on $\beta$-propellers, which is a limitation to state in the thesis rather than a fact about neuraminidase.

Either way it costs one GPU-minute and it changes what we would write.

### 10.8 What this means for the decision

The explanations differ scientifically but not practically: for the purpose of choosing a target, neuraminidase is out either way, because the model's neuraminidase is not the real one ($E \approx 0.6$). The diagnosis is still worth an afternoon, because "our decoder cannot close a $\beta$-propeller" — or "cannot fold a subunit of an obligate oligomer" — is a statement about the method's domain of validity, and that belongs in the thesis.

## 11 What the energy is actually measured against

This is the subtlety most worth being explicit about tomorrow, because it decides whether "use neuraminidase anyway" is coherent.

In the paper, equation (1) compares the predicted structure of a candidate sequence with a reference conformation $r_0$. In *this* implementation the energy that the Monte Carlo minimises is

$$U_{\mathrm{am}}(\sigma) = \sum_{i<j} \Big[ \log A_{ij}(\sigma) - \log A_{ij}(\sigma_{\mathrm{ref}}) \Big]^2, \tag{2}$$

where $A(\sigma)$ is the symmetrised final-layer attention map that ESM3 produces for sequence $\sigma$ (`utils/energies.py:compute_U_am`, `custom_esm/models/esm3.py:predict_attention`). Three consequences, none of them obvious from the paper:

1. **No crystal structure enters the energy at all.** The reference is the model's own output for the reference *sequence*. This is why $U(\sigma_{\mathrm{ref}}) = 0$ holds exactly, a property the code relies on as a sanity check, and why the landscape is well posed whatever the experimental structure looks like.
2. **The energy is differentiable in the sequence probabilities**, which is what makes the gradient-informed move of `one_pass_move.tex` possible at all. A discrete contact map is not. The optional second term $U_{\mathrm{structure_ce}}$ is closer in spirit to the paper's contact-map energy and is also differentiable, but it is off by default and has no exact zero at the reference.
3. **So the decode gate is not a test of well-posedness.** It is a test of *interpretation*: whether a low $U$ can be read as "this sequence folds like the real protein".

Hence two defensible positions, which the supervisor should pick between explicitly:

**(i) The reference must be a protein the model predicts correctly.** Then low $U$ means "folds like the real thing", the per-site entropy comparison against natural sequences tests something clean, and biological conclusions are available. Neuraminidase is excluded. This is the position the gate implements.

**(ii) The reference need only be well defined.** Then neuraminidase, or any sequence, is a usable *benchmark*: acceptance rates, decorrelation times, the speed-up from the one-pass move and the $n_{\text{sites}}$ scan are all measured exactly as before, and the thesis's stated aim — making the method faster — is served. What is lost is biology: comparing the resulting entropies with a natural alignment no longer tests anything, because the reference is not the natural fold.

The paper did not have to choose: protein G is small and predicted well, so the two readings coincide. We now have a target where they do not, which is why the choice should be made out loud and stated with every figure.

## 12 Cytochrome P450cam passes the gate

Run on astaroth on 8 November 2026 (`compare_structure_p450cam.log`), P450cam, UniProt P00183, 414 residues, against PDB 2CPP, 32 generation steps:

| | P450cam / 2CPP | NA head / 2HU0 |
|---|---:|---:|
| length decoded | 414 aa | 385 aa |
| residues paired with the crystal | 405 (98%) | 385 (100%) |
| mean pLDDT | **91.9** | 67.7 |
| residues with pLDDT $\geq 90$ | 80% | 9% |
| residues with pLDDT $< 50$ | **0%** | 17% |
| $R_g$ predicted / experimental | 21.70 / — | 19.72 / 18.88 |
| C$\alpha$ RMSD, all paired | **1.98 Å** | 14.93 Å |
| core RMSD / residues | 0.96 Å / 320 (79%) | 1.17 Å / 116 (30%) |
| residues within 2 Å | 77% | 11% |
| TM-score (lower bound) | **0.942** | 0.563 |
| crystal contacts recovered | **88%** | 39% |
| $E$ of eq. (1) | **0.109** | 0.596 |

This is a different kind of result, not a better score on the same scale. Four fifths of the chain superposes at sub-angstrom accuracy, no residue is predicted with low confidence, and the worst single deviation in the whole protein is 7.6 Å — a surface loop around residues 186–192, plus the last two residues of the chain, which are a terminus. The nine N-terminal residues of our construct are absent from the crystal and are simply left out of every number. For comparison, neuraminidase's worst deviations are 40–50 Å.

Two caveats to state when quoting these numbers. 2CPP carries camphor and a heme group; the scripts ignore non-protein atoms, so a few active-site residues are held by a ligand in the crystal that the ligand-free prediction cannot reproduce, which is the likely origin of part of the residual 1.98 Å. And P450cam has no disulfide bridges, so that diagnostic is silent here — it reports zero, which is correct and not a failure.

### 12.1 A measured gate, instead of a guessed one

We now have the calibration that was missing when the gate was first written:

| | P450cam (passes) | NanH (control) | protein G (the paper's) | NA head (fails) | HA0 (fails) |
|---|---|---|---|---|---|
| length | 414 aa | 381 aa | 56 aa | 385 aa | 566 aa |
| fold | $\alpha/\beta$ | 6-blade | $\alpha/\beta$ | 6-blade | — |
| mean pLDDT | 0.92 | 0.86 | 0.92 | 0.68 | 0.51–0.62 |
| C$\alpha$ RMSD, all | 1.98 Å | 3.20 Å | 3.02 Å | 14.93 Å | — |
| core RMSD / residues | 0.96 / 79% | 0.94 / 73% | 0.70 / 86% | 1.17 / 30% | — |
| TM-score | 0.942 | 0.895 | 0.821 | 0.563 | — |
| $E$ against the crystal | 0.109 | 0.230 | 0.180 | 0.596 | not measured |
| crystal contacts recovered | 88% | 76% | 80% | 39% | not measured |

Two lessons, both of which argue against a single sharp threshold.

**Protein G: short chains sit higher.** Protein G is the target of Zambon et al. (2024) and ESM3 predicts it well by every local measure — pLDDT 0.92, a 0.70 Å core over 86% of the chain — yet its $E$ is 0.180, higher than P450cam's 0.109, and its global RMSD is 3.0 Å. At 56 residues there are only 137 contacts, so a frayed terminus moves $E$ a great deal, whereas P450cam's 1195 contacts average that kind of local error away.

**NanH: a correct fold can sit at $E = 0.23$.** The sialidase control is a genuinely correct prediction — TM 0.895, three quarters of the chain inside 1 Å, the propeller closed — and still reaches $E = 0.230$ at 381 residues, where length cannot be the excuse. Its error is concentrated in the first thirty residues (6.0 and 6.4 Å for the first two blocks, against 0.7–2 Å through the middle and 0.83 Å at the C-terminal block), i.e. one misplaced N-terminal segment is enough to double $E$ while leaving the fold right.

**So the gate, restated.** "Good" is $E \approx 0.1$–$0.25$ and "failed" is $E \gtrsim 0.5$; the gap is still a factor of two and nothing we have measured falls inside it. But $E$ alone is not the criterion: it must be read together with the pLDDT, the *fraction* of the chain in the core, and the TM-score, because those three separate "a correct fold with one loose end" (NanH, 73% of the chain at 0.94 Å) from "a third of a fold" (NA, 30% at 1.17 Å). A single number would have called the paper's own protein marginal and the sialidase control a failure.

$E$ for HA0 is still worth computing for the thesis, which needs the 2HMG entry downloaded on astaroth; `ha0_pred.pdb` is already in the repository.

### 12.2 Step-count stability for the chosen target

The gate includes repeating the decode at several step counts, since that is how HA0 and NA were convicted. For P450cam (8 November):

| steps | pLDDT | $R_g$ (Å) | RMSD all (Å) | core / residues | TM | $E$ |
|---|---|---|---|---|---|---|
| 8 | 90.3 | 21.68 | 1.99 | 0.94 Å / 318 (79%) | 0.942 | 0.106 |
| 32 | 91.9 | 21.70 | 1.98 | 0.96 Å / 320 (79%) | 0.942 | 0.109 |
| 128 | 94.2 | 21.88 | 2.36 | 0.95 Å / 315 (78%) | 0.938 | 0.111 |

Flat, as it should be: $R_g$ within 0.2 Å, the core fixed at 0.94–0.96 Å over 78–79% of the chain, $E$ within 0.005. The same mild pattern as NA appears — more steps raise the confidence (90.3 to 94.2) and very slightly worsen the agreement (RMSD 1.99 to 2.36 Å) — but here it is a tenth of an Ångstrom rather than a different structure. 32 steps is a sensible default to keep.

## 13 The control: a monomeric propeller folds correctly

This is the experiment proposed in section 10.7 and it was run the same evening, so the "maybe ESM3 is just bad at $\beta$-propellers" explanation can be removed from the discussion.

Bacterial sialidases have the same six-bladed propeller fold as viral neuraminidase but are monomeric; the sialidase of *Salmonella typhimurium* (NanH, PDB 2SIL, 381 residues) was solved precisely to demonstrate the shared fold (Crennell et al., 1993). Its sequence was taken straight from the crystal with `pdb_to_reference.py`, so no accession lookup was needed.

| | NanH / 2SIL<br>monomeric propeller | NA head / 2HU0<br>tetrameric propeller |
|---|---:|---:|
| length | 381 aa | 385 aa |
| mean pLDDT | **86.3** | 67.7 |
| residues with pLDDT $< 50$ | **0%** | 17% |
| C$\alpha$ RMSD, all paired | **3.20 Å** | 14.93 Å |
| core RMSD / residues | 0.94 Å / 277 (**73%**) | 1.17 Å / 116 (30%) |
| TM-score | **0.895** | 0.563 |
| $E$ against the crystal | **0.230** | 0.596 |
| first block (residues 1–20) | 6.0 Å | 29.6 Å |
| last block (C-terminal 20) | **0.83 Å** | 21–41 Å |
| long-range disulfide | **1 of 1 formed** | 0 of 1 formed |

The last three rows are the ones that matter, and they are why this control was worth a GPU-minute:

- **The ring closes.** NanH's C-terminal twenty residues land at 0.83 Å from their crystal positions. In a propeller that is the strand which completes the first blade, so the model demonstrably can make the molecular Velcro contact. In NA the same region is 21–41 Å out.
- **Its one long-range disulfide is formed**: Cys42–Cys103, 61 residues apart, 4.9 Å predicted against 4.25 Å in the crystal. NA's Cys92–Cys417, 325 residues apart, is 20.5 Å.
- **The remaining error is one segment, not the architecture.** NanH's first thirty residues are 6 Å out and everything from residue 41 to the end is 0.7–5 Å. That is a loose N-terminal extension on a correct fold.

**What this excludes, and what it leaves.** ESM3-open can fold a six-bladed $\beta$-propeller from sequence alone, closed ring and all. So the failure on neuraminidase is *not* a property of the fold class, and section 10.7 is settled negatively. What remains is what distinguishes the two proteins:

- NanH is monomeric; NA is an obligate homotetramer with a $Ca^{2+}$ ion on its fourfold axis (section 10, the quaternary explanation, which is now the leading one).
- NA carries eight disulfide bridges to NanH's one, and is glycosylated, while NanH is a bacterial cytoplasmic/secreted protein.
- NA's ring closure is stapled by a 325-residue-separation disulfide, a much longer-range constraint than anything in NanH.

These are not mutually exclusive, and the disulfide-prompting experiment of section 10 would separate the second from the first.

**One honest caveat.** NanH is not a control for *viral* neuraminidase in the strict sense: it shares the fold but not the sequence family (the structures were solved to show that distant homology). So it proves that the architecture is predictable, not that NA's particular sequence should have been.

# 14 The pipeline from here, and its expected cost

With a target that passes, this is what the following weeks look like. The numbers are extrapolations from the HA runs, so they are orders of magnitude, not promises.

1. **Sanity checks on the new reference** (minutes). $U(\sigma_{\text{ref}}) = 0$ exactly, by equation (2), and the single-mutant $\Delta U$ distribution from `scan_U_am_landscape.py` should contain no negative values. These reproduce known protein G landmarks and catch a broken harness before any long run.

2. **Temperature scan** (hours). The selective temperature is set by the peak of $\mathrm{Var}(U)/T^2$. For HA the transition sat around $T \approx 8$–$18$, with $T \approx 5$ inside the reference basin and $T \gtrsim 32$ unfolded. A 414-residue protein with a different fold will have its own scale: this must be measured, not carried over.

3. **Equilibration** (days). At 566 residues the one-pass move cost 0.153 s per move, a 2.4× speed-up over site-soft. Scaling with length, expect roughly 0.10–0.12 s per move at 414 residues, about $10^5$ moves per three hours. HA needed of order $10^6$ moves to plateau at $T = 12$ and had *not* equilibrated at $T = 5$ after 580k, which is exactly the problem this thesis is about. Budget one to three days per low-temperature chain.

4. **The $n_{\text{sites}}$ scan** (a day). At HA the answer was $N = 1$ at low temperature, with acceptance falling roughly as $a_1^N$. Unless the proposal itself changes, expect the same; a different answer would be interesting in itself.

5. **The K-site comparison** (days, mostly data work). `ha_site_entropy.py` takes any aligned set, so it can build the natural per-site entropy for the new family. Report $N_{\text{eff}}$ per column after the redundancy weighting rather than raw sequence counts: for P450 the family is deep enough that $N_{\text{eff}}$ should stay in the thousands, which is the main reason to prefer it over a viral family, where the 0.8 weighting collapses each subtype to about one cluster.

6. **Structural check on the sampled ensemble** (hours). Decode a handful of sampled sequences and report RMSD and $E$ against the reference: this is what tells us whether low energy really means "folds like the reference". With P450cam's $E = 0.11$ floor in hand, that statement now has a scale.

## 14.1 What we can say tomorrow, and what we cannot

We can say that neuraminidase does not pass, with numbers, and that the failure is reproducible, specific, and stable across step counts. We can say that the replacement is found and measured, including its step-count stability: P450cam, 414 residues, pLDDT 90–94, RMSD 2 Å, TM 0.94, $E = 0.11$. We can say what the gate is and why it cannot be a single number. And we can say that the fold class is not to blame, because a monomeric protein with the same propeller architecture is predicted correctly, ring closed and long-range disulfide formed.

What we cannot say is which NA-specific cause is responsible. The tetramer is now the leading explanation, the disulfide load and the glycans are alternatives, and the three are not mutually exclusive. Two experiments would narrow it: the disulfide-prompted decode (an hour,

tooling already here) and a multimer predictor on the tetramer (a week, new tooling). Neither changes the target decision.

### 14.2 Questions for the supervisor

1. Section 11: should the reference always be a protein the model predicts correctly (position (i)), or is a well-defined but biologically-wrong reference acceptable for the purely methodological part of the thesis (position (ii))?

2. Is the gate right as a criterion: pLDDT $\gtrsim 0.9$ *and* an $E$ against the crystal well below 0.3, read with the chain length in mind, as section 12.1 argues from protein G?

3. P450cam as the new target — or P450-BM3's heme domain, which is ∼455 residues and from the same family, or luciferase at 550 residues, which also passes the decode gate but has a more divergent family?

4. The sialidase control is done (section 13) and the tetramer is now the leading explanation for NA. Is it worth a week of new tooling — a multimer predictor — to confirm that, or is it enough for the thesis to record that the subunit of an obligate oligomer was not predictable and move on?

5. Does he want the NA post-mortem written up at all, or is it enough to record that the target was changed?

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